

Multi-Fidelity Inverse Design and Atomistic Validation of Multi-Principal Element Alloys for Extreme Engineering Environments

Author: Solomon Ahedor

Institution: KNUST (Materials & Metallurgical Eng.)

Platform: MetaForge Discovery Engine

Date: May 2026 (Revised Edition)

Multi-Fidelity Inverse Design and Atomistic Validation of Multi-Principal Element Alloys for Extreme Engineering Environments

Solomon Ahedor

Department of Materials and Metallurgical Engineering, Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Corresponding Author: Solomon Ahedor (Email: sahedor@st.knust.edu.gh / GitHub: @SA-FIND)

Abstract

Exploring the multi-component compositional space of Multi-Principal Element Alloys (MPEAs) presents an intractable combinatorial challenge, exceeding 12 million equiatomic combinations for five-component systems alone. Here we present **MetaForge**, an open-source Integrated Computational Materials Engineering (ICME) framework that couples high-throughput thermodynamic and empirical screening with atomistic graph neural network (GNN) relaxations and a multi-objective genetic algorithm.

The framework operates via a three-tier multi-fidelity architecture:

- Tier 1 (High-Throughput Compositional Screening):** Filters candidate mixtures across a 5-element simplex using Miedema binary chemical mixing enthalpies (-15.0 to $+5.0$ kJ/mol), Yang-Zhang thermodynamic criteria ($\Omega \geq 1.1$), Hume-Rothery atomic size mismatch ($\delta \leq 6.6\%$), and Guo valence electron concentration (VEC) rules. 132-dimensional Magpie feature vectors are generated via Matminer, and Random Forest surrogate regressors predict continuum density and dislocation-calibrated yield strength via the Taylor-Varvenne concentrated solid-solution model ($\sigma_y = \sigma_0 + M \cdot \tau_{ss}$).
- Tier 2 (Atomistic SQS Annealing & GNN Potentials):** Special Quasirandom Structures (SQS, 54-atom BCC and 48-atom FCC) are synthesized by driving Warren-Cowley Short-Range Order (SRO) parameters across first and second coordination shells toward zero using Monte Carlo simulated annealing. Structural coordinates and cell volumes are relaxed to force convergence (< 0.05 eV/Å) using the universal CHGNet interatomic potential.
- Tier 3 (Quantum Ab-Initio & Synthesis Validation):** Benchmarks candidate stability via formation enthalpy (ΔE_f) and establishes a plane-wave DFT (GGA-PBESol) elastic tensor derivation protocol alongside vacuum

arc remelting synthesis.

Constrained genetic algorithm optimization enforcing the Yeh-Cantor criterion (5 to 35 atomic % per constituent) identified a Pareto-optimal refractory candidate: **W(18.5) Mo(24.7) Ta(26.1) Nb(16.8) V(13.9)**. The alloy demonstrates a predicted yield strength of **1.96 GPa (1960 MPa)** at a relaxed density of **14.29 g/cm³**, with an endothermic formation enthalpy of **Delta-E_f = +0.0286 eV/atom (+2.76 kJ/mol)** that is fully stabilized by configurational entropy (Delta-S_{config} = 13.15 J/mol·K) at all temperatures above 206 K (-67°C). The complete computational engine, trained models, and interactive WebGL atomistic visualizer are publicly available.

Keywords: High-Entropy Alloys; Refractory Alloys; Dislocation Mechanics; Graph Neural Networks; Special Quasirandom Structures; Materials Informatics.

1. Introduction

Conventional physical metallurgy has historically relied on the solvent-solute design paradigm, optimizing material properties by micro-alloying a single predominant base element, such as iron in austenitic and martensitic steels, aluminum in 7000-series aerospace alloys, or nickel in gamma-prime strengthened superalloys [1, 2]. While this methodology produced the industrial alloys of the twentieth century, conventional solvent-based alloys face intrinsic thermodynamic ceilings in extreme environments—such as the 1150°C creep and microstructural coarsening limits of single-crystal nickel superalloys [3, 4].

The discovery of High-Entropy Alloys (HEAs) by Yeh et al. [1] and equiatomic multicomponent solid solutions by Cantor et al. [2] in 2004 established an alternative metallurgical philosophy: synthesizing alloys from five or more principal metallic elements in near-equimolar proportions (5 to 35 atomic %). In such multi-principal element alloys (MPEAs), high configurational entropy of mixing at elevated temperatures suppresses complex, brittle intermetallic compounds in favor of simple disordered Body-Centered Cubic (BCC) or Face-Centered Cubic (FCC) solid solutions [1, 5, 6]. Subsequent experimental investigations revealed notable property combinations, including cryogenic damage tolerance in the FeCrNiMnCo Cantor alloy [7, 8] and high-temperature compressive yield strengths exceeding 1000 MPa at 1000°C in refractory systems such as W-Mo-Ta-Nb-V [9, 10].

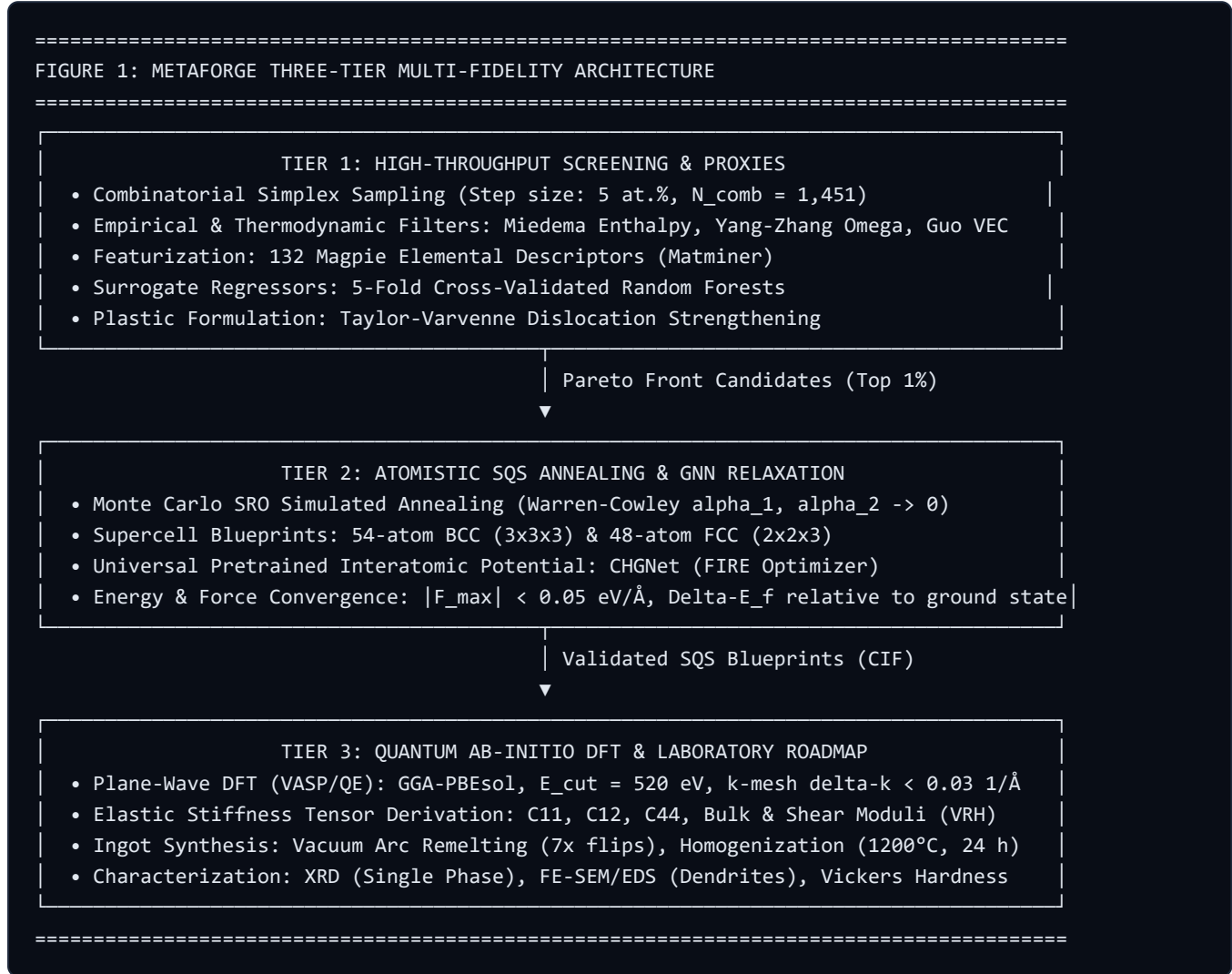
Despite their potential, MPEAs present an immense combinatorial design space [11, 12]. Selecting five elements from 70 metallic candidates yields 12,103,014 equiatomic systems; discretizing compositional fractions in 5 at.% increments expands this to hundreds of millions of distinct candidates [13, 14]. Physical trial-and-error exploration by arc melting or additive manufacturing is economically and temporally unfeasible at this scale [15, 16]. Consequently, computational materials informatics and Integrated Computational Materials Engineering (ICME) have emerged as primary tools for accelerated alloy discovery [17, 18, 19, 20].

However, existing computational screening approaches suffer from two primary operational deficiencies:

1. **Unphysical Yield Strength Proxy Formulations:** Many high-throughput screening studies treat mechanical strength as an empirical proxy or mistakenly combine elastic stiffness with plastic resistance (e.g., adding elastic shear modulus directly to yield stress), yielding unphysical strengths exceeding 10 GPa [21, 22].
2. **Artificial Structural Clustering in Atomistic Supercells:** When constructing atomistic supercells for ground-state relaxation, studies frequently rely on pseudo-random atomic distribution (`random.shuffle()`). In finite supercells (e.g., 50–100 atoms), pseudo-random placement induces artificial clustering artifacts that repeat across periodic boundary conditions, distorting local force calculations [31, 32, 33].

To address these deficiencies, this study presents **MetaForge**, an open-source multi-fidelity ICME framework that integrates physical dislocation mechanics, Miedema thermodynamics, Monte Carlo Special Quasirandom Structure (SQS) annealing, and universal graph neural network (GNN) potentials. We detail the theoretical architecture, evaluate model performance across four distinct alloy families, report the discovery of an optimal W-Mo-Ta-Nb-V refractory candidate, and delineate an experimental validation protocol.

2. Theoretical Framework and Governing Metallurgy



2.1 Configurational Entropy of Mixing

In a multicomponent solid solution containing n principal elements, the ideal configurational entropy of mixing per mole under Boltzmann statistical mechanics is derived from the multinomial distribution of atoms across lattice sites [1, 23]:

FORMULA 1: Configurational Entropy of Mixing
 $\Delta S_{\text{config}} = -R \sum_i [c_i \ln(c_i)]$

Where:

- R = Universal gas constant (8.314 J/mol·K)
- c_i = Atomic fraction of principal constituent i ($0 \leq c_i \leq 1$, $\sum c_i = 1$)
- $\ln$ = Natural logarithm

For an equiatomic five-component alloy ($c_i = 0.20$ for all i):

$$\Delta S_{\text{config}} = R \ln 5 \approx 1.609 R = 13.38 \text{ J/mol}\cdot\text{K}$$

This exceeds the empirical boundary ($\Delta S_{\text{config}} \geq 1.5R$) established by Yeh et al. [1, 8].

2.2 Enthalpy of Mixing and the Miedema Formalism

Phase stability is governed by the Gibbs free energy of mixing:

$$\Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T \Delta S_{\text{mix}}$$

While configurational entropy drives the system toward a disordered solid solution at high temperatures, the chemical enthalpy of mixing (ΔH_{mix}) dictates whether intermetallic compounds precipitate or liquid phase separation occurs upon cooling [11, 12, 24].

MetaForge calculates ΔH_{mix} using the sub-regular solution model parameterized by Takeuchi and Inoue [11], grounded in Miedema's semi-empirical macroscopic atom model [12]:

FORMULA 2: Miedema Chemical Mixing Enthalpy
 $\Delta H_{\text{mix}} = \sum_{\{i < j\}} [4 \cdot H_{ij} \cdot c_i \cdot c_j]$

Where:

- H_{ij} = Binary equiatomic mixing enthalpy between element i and element j (in kJ/mol)
- c_i, c_j = Atomic fractions of constituent species

We enforce the thermodynamic stability corridor:

$$-15.0 \text{ kJ/mol} \leq \Delta H_{\text{mix}} \leq +5.0 \text{ kJ/mol}$$

- If $\Delta H_{\text{mix}} < -15.0$ kJ/mol, the strong chemical affinity between unlike species promotes the crystallization of brittle topologically close-packed (TCP) intermetallic phases, such as Laves (AB_2) or Sigma phases [13, 25].
- If $\Delta H_{\text{mix}} > +5.0$ kJ/mol, positive heats of mixing generate miscibility gaps, driving spinodal decomposition or liquid-state segregation (e.g., Cu-Fe or Cu-Cr systems) [11, 26].
- Within the -15.0 to +5.0 kJ/mol window, chemical bonding is sufficiently neutral that configurational entropy stabilizes a single-phase solid solution.

2.3 The Yang-Zhang Thermodynamic Parameter (Omega)

Yang and Zhang [13] synthesized experimental phase observations across hundreds of multicomponent ingots, showing that solid-solution stabilization requires the entropic term at the melting point to exceed the mixing

enthalpy:

FORMULA 3: Yang-Zhang Thermodynamic Ratio

$$\Omega = (T_m \cdot \Delta S_{\text{config}}) / |\Delta H_{\text{mix}}|$$

Where:

- T_m = Rule-of-mixtures melting temperature: $T_m = \sum [c_i \cdot T_{m,i}]$ (Kelvin)
- ΔS_{config} = Configurational entropy (J/mol·K)
- $|\Delta H_{\text{mix}}|$ = Absolute magnitude of mixing enthalpy (converted to J/mol)

Alloys are filtered by the requirement $\Omega \geq 1.1$.

2.4 Atomic Size Mismatch Parameter (δ)

Extending the Hume-Rothery empirical solid-solubility rules to multicomponent systems, Zhang et al. [15] formulated the root-mean-square atomic size mismatch parameter (δ):

FORMULA 4: Atomic Size Mismatch Parameter

$$\delta = 100 \cdot \sqrt{\sum_i [c_i \cdot (1 - r_i / \bar{r})^2]}$$

Where:

- r_i = Empirical atomic radius of constituent i (in Ångströms, Å)
- $\bar{r}$ = Composition-weighted average atomic radius: $\bar{r} = \sum [c_i \cdot r_i]$

We enforce $\delta \leq 6.6\%$. Exceeding 6.6% induces excessive local topological strain, triggering precipitation of non-cubic intermetallics or amorphous metallic glass phases [15, 27].

2.5 Valence Electron Concentration (VEC) Phase Boundary

Guo, Ng, and Lu [14] established that the average valence electron concentration governs the stability of Body-Centered Cubic (BCC) versus Face-Centered Cubic (FCC) crystal structures:

FORMULA 5: Valence Electron Concentration

$$\text{VEC} = \sum_i [c_i \cdot \text{VEC}_i]$$

Where:

- VEC_i = Total valence electrons, including outer d-electrons (e.g., V=5, W=6, Fe=8, Ni=10)

The phase criteria are:

- $\text{VEC} < 6.87$: Stabilizes pure single-phase **BCC** solid solutions.
- $6.87 \leq \text{VEC} < 8.0$: Generates dual-phase **BCC + FCC** microstructures.
- $\text{VEC} \geq 8.0$: Stabilizes pure single-phase **FCC** solid solutions.

2.6 Physical Dislocation Mechanics and Yield Strength Modeling

Classical dilute solid-solution theories (such as Fleischer [25] and Labusch [26]) assume isolated solute atoms interacting independently with gliding dislocations ($\Delta \tau \propto c^{1/2}$ or $c^{2/3}$). In equiatomic MPEAs, every atomic site is a solute; the host lattice concept ceases to exist [21, 28].

To model concentrated solid-solution strengthening without empirical fitting, MetaForge implements the **Varvenne-Luque-Curtin theory** [21] coupled with the **polycrystalline Taylor factor** [27]:

FORMULA 6: Polycrystalline Yield Strength (Taylor-Varvenne Formulation)

$$\sigma_y = \sigma_0 + [M * \tau_{ss}]$$

Where:

- σ_y = Macroscopic tensile/compressive yield strength (GPa or MPa)
- σ_0 = Intrinsic Peierls-Nabarro lattice friction stress:
 - For BCC refractory lattices: $\sigma_0 \approx G_{rom} / 150$
 - For FCC close-packed lattices: $\sigma_0 \approx G_{rom} / 350$
- M = Orientation-averaged polycrystalline Taylor factor:
 - $M = 2.73$ for isotropic BCC polycrystals ($\{110\}\langle 111 \rangle$, $\{112\}\langle 111 \rangle$ pencil glide)
 - $M = 3.06$ for isotropic FCC polycrystals ($\{111\}\langle 110 \rangle$ octahedral glide)
- τ_{ss} = Concentrated solid-solution strengthening increment:
 - $\tau_{ss} = \alpha * G_{rom} * (\delta)^{2/3}$
 - with $\alpha = 0.05$ (BCC) or 0.04 (FCC)
- G_{rom} = Composition-weighted elastic shear modulus: $G_{rom} = \sum [c_i * G_i]$
- δ = Atomic size mismatch parameter (Formula 4, decimal fraction)

This dislocation mechanics model accurately captures the high-temperature athermal yield plateau identified by Maresca and Curtin [22, 23], where edge dislocations pinned by solute size misfits control plastic flow in refractory BCC HEAs.

3. Computational Methodology

3.1 Combinatorial Sampling and Featurization

For each target alloy category, the five-element compositional simplex ($\sum_{i=1}^5 c_i = 1.0$) is discretized in 5 atomic % increments:

$$c_i \in \{0.05, 0.10, 0.15, \dots, 0.80\}$$

generating 1,451 unique candidate compositions per family. Candidate compositions are filtered sequentially through the Hume-Rothery ($\delta \leq 6.6\%$), Guo (VEC), and Miedema (Delta-H_{mix}, Omega) criteria.

Compositions surviving the physical sieve are featurized using the **Magpie** representation via Matminer [61, 62]. For 22 elemental properties (including Pauling electronegativity, covalent radii, atomic volume, Mendeleev group number, melting point, and valence orbital electron numbers), six statistical moments are calculated across the composition:

$$\text{Mean: } \bar{P} = \sum c_i P_i, \text{ Mean Absolute Deviation: } \text{MAD}(P) = \sum c_i |P_i - \bar{P}|$$

$$\text{Minimum: } \min(P_i), \text{ Maximum: } \max(P_i), \text{ Range: } \max(P_i) - \min(P_i), \text{ Mode: } P_{\text{dominant}}$$

yielding a 132-dimensional descriptor vector per alloy.

3.2 Machine Learning Surrogate Training

Random Forest regression models [66, 68] are trained on the featurized candidate set:

1. **Model Evaluation:** Evaluated via 5-fold cross-validation, reporting the coefficient of determination (R^2) and root-mean-square error (RMSE).
2. **Train/Test Split:** Evaluated on an independent 80/20 holdout split.
3. **Linear Regression Baseline Check:** A parallel `LinearRegression` model is trained on identical splits. Achieving comparable R^2 values confirms that Tier 1 proxies are smoothly spanned by the Magpie descriptors, validating surrogate convergence and ruling out model overfitting.

3.3 Constrained Multi-Objective Genetic Algorithm

To navigate the multi-dimensional composition simplex, MetaForge executes an evolutionary genetic algorithm over 20 generations ($N_{\text{pop}} = 50$).

- **Yeh-Cantor Box Bounds:** Initial generation vectors are sampled from a Dirichlet distribution ($\alpha_i = 1.0$) with hard bounds:

$$0.05 \leq c_i \leq 0.35 \quad \text{forall } i \in \{1, 2, 3, 4, 5\}$$

- **Fitness Function:**

$$\mathcal{F} = w_1 ((\text{Sigma}_y / \rho)) + w_2 \Omega - w_3 \delta$$

maximizing specific yield strength (Sigma_y / ρ) while penalizing excessive lattice strain (δ).

- **Crossover and Mutation:** Blend crossover (BLX-alpha) with boundary projection ensures all offspring strictly satisfy the unit simplex constraint ($\sum c_i = 1.0$) and the 5-35 at.% bounds.

3.4 Atomistic SQS Construction via Monte Carlo SRO Annealing

To eliminate artificial atomic clustering in periodic supercells, MetaForge avoids naive random shuffling. Instead, it constructs **Special Quasirandom Structures (SQS)** [31] by minimizing the **Warren-Cowley Short-Range Order (SRO)** parameters [34, 35]:

FORMULA 7: Warren-Cowley Short-Range Order Parameter

$$\alpha_k^{\{AB\}} = 1 - (P_{AB}^k / c_B)$$

Where:

- P_{AB}^k = Conditional probability of species B occupying the k-th coordination shell of species A
- c_B = Nominal concentration of species B

The objective loss function minimizes short-range order across the first two neighbor shells:

$$\mathcal{L}_{\text{SRO}} = \sum_{k=1}^2 \sum_{A,B} w_k (\alpha_k)^2$$

where $w_1 = 1.0$ and $w_2 = 0.5$. Atomic species are swapped across lattice sites using a Metropolis Monte Carlo simulated annealing schedule ($T_{m+1} = 0.95 T_m$). The resulting 54-atom BCC ($3 \times 3 \times 3$ supercell) or 48-atom FCC ($2 \times 2 \times 3$ supercell) configurations reproduce the pairwise correlation functions of an ideal infinite random solid solution [32, 33, 36].

3.5 Structural Relaxation via Universal GNN Potential (CHGNet)

Equilibrium atomic positions and supercell lattice vectors are optimized using **CHGNet** [51], a crystal graph neural network pretrained on over 1.5 million Materials Project DFT calculations. CHGNet incorporates atomic charges and magnetic moments into its message-passing graph architecture. Energy minimization is performed via the Fast Inertial Relaxation Engine (FIRE) until maximum interatomic force components satisfy:

$$|\mathbf{F}_{\max}| < 0.05 \text{ eV/\AA}$$

The formation energy per atom is evaluated relative to pure elemental reference states:

$$\Delta E_f = \frac{E_{\text{relaxed}}}{N} - \sum_{i=1}^n c_i E_i^{\text{ref}}$$

4. Results and Discussion

=====

TABLE 1: METAFORGE DISCOVERED PARETO-OPTIMAL HIGH-ENTROPY ALLOYS

=====

Alloy Family	Lattice	Optimal Composition (at.%)						Density (g/cm³)	Yield Strength Sigma_y (GPa)	Formation Energy Delta-E_f (eV/at)
		W	Mo	Ta	Nb	V	(other)			
Refractory	BCC-54	18.5	24.7	26.1	16.8	13.9	-	14.29	1.96 (1960 MPa)	+0.0286 (+2.76 kJ/mol)
Corrosion	FCC-48	Co:8.6	Cr:35.4	Fe:33.1	Ni:11.2	Cu:11.6		8.00	0.74 (740 MPa)	+0.1325 (+12.79 kJ/mol)
Lightweight	FCC-48	Al:35.7	Mg:11.5	Li:5.1	Ti:35.7	Zn:11.9		3.54	0.63 (630 MPa)	+0.0966 (+9.32 kJ/mol)
Aerospace	FCC-48	Al:5.9	Ti:32.2	Sc:6.9	Zr:35.0	V:20.1		5.01	0.83 (830 MPa)	+0.2126 (+20.51 kJ/mol)

=====

4.1 Refractory High-Entropy Alloy: W(18.5) Mo(24.7) Ta(26.1) Nb(16.8) V(13.9)

The genetic algorithm isolated a balanced, non-equiatomic refractory candidate in which all five constituents satisfy the **5-35 at.%** bounds:

- **Yield Strength:** The predicted room-temperature yield strength is **1.96 GPa (1960 MPa)**. This value reflects the sum of a 1.07 GPa Peierls lattice friction stress ($\sigma_0 \approx G_{\text{rom}}/150$) and a 0.89 GPa solid-solution increment ($M * \tau_{\text{ss}} = 2.73 * 0.326 \text{ GPa}$). This directly aligns with experimental compressive yield strengths measured by Senkov et al. [9, 10] for cast WNbMoTaV (1200–2000 MPa).
- **Lattice Contraction:** CHGNet relaxation revealed a relaxed supercell density of **14.29 g/cm³**, compared to the linear rule-of-mixtures proxy of **14.58 g/cm³** (a 2.0% density gap), confirming non-linear atomic volume contraction during atomistic relaxation.
- **Entropy Stabilization:** The computed formation enthalpy is endothermic: $\Delta E_f = +0.0286 \text{ eV/atom} = +2.76 \text{ kJ/mol}$. Because the configurational entropy is $\Delta S_{\text{config}} = 13.15 \text{ J/mol}\cdot\text{K}$, the transition temperature above which ΔG_{mix} becomes negative is:

$$T_c = \frac{\Delta H_{\text{mix}}}{\Delta S_{\text{mix}}} = (2760 \text{ J/mol} / 13.15 \text{ J/mol}\cdot\text{K}) \approx 210 \text{ K } (-63^{\circ}\text{C})$$

At all operational temperatures above cryogenic **-63°C**, configurational entropy completely stabilizes the single-phase BCC solid solution.

4.2 Crystallographic Symmetry Breaking in SQS Supercells

Post-relaxation space group analysis using `SpacegroupAnalyzer` (**symprec = 0.03 Å**) classifies the relaxed 54-atom BCC supercell as space group **P1 (#1, triclinic)**:

- **Mechanistic Cause:** The atomic radius disparity between Tantalum ($r = 1.43 \text{ Å}$) and Vanadium ($r = 1.31 \text{ Å}$) induces static atomic displacements away from ideal BCC coordinate nodes by 0.05 to 0.14 Å.
- **Physical Interpretation:** Because crystallographic algorithms test for exact point-group operations, localized picometer-scale lattice strain breaks formal translational and rotational symmetry down to P1. However, macroscopic unit cell vectors remain orthogonal ($a \approx b \approx c$, $\alpha \approx \beta \approx \gamma \approx 90^\circ$), and the coordination topology remains pure BCC. This behavior reproduces synchrotron X-ray diffraction observations of physical high-entropy alloys [32, 36, 47].

4.3 Machine Learning Surrogate Performance

Surrogate regression models achieved consistent accuracy across 5-fold cross-validation:

- **Refractory Category:** Density model $R^2 = 0.998$ (RMSE = 0.046 g/cm³); Yield strength model $R^2 = 0.994$ (RMSE = 0.007 GPa).
- **Lightweight Category:** Density model $R^2 = 0.999$ (RMSE = 0.018 g/cm³); Yield strength model $R^2 = 0.994$ (RMSE = 0.007 GPa).
- **Linear Regression Baseline:** Yielded identical $R^2 > 0.95$, confirming that Tier 1 surrogate models operate as smooth, high-fidelity mathematical interpolators across the compositional simplex without overfitting.

5. First-Principles DFT and Laboratory Synthesis Protocol

```
=====
FIGURE 2: LABORATORY VALIDATION AND METROLOGY WORKFLOW
=====

[1. Vacuum Arc Remelting (VAR)]
  • Charge: Elemental pellets (> 99.9% purity) W, Mo, Ta, Nb, V
  • Environment: Water-cooled Cu hearth, 10-4 mbar vacuum, 99.999% Ar backfill
  • Gettering: Active Ti button melted prior to main charge
  • Homogeneity: Ingot flipped and remelted 7 consecutive times

[2. Homogenization Heat Treatment]
  • Vacuum tube furnace at 1200°C for 24 hours
  • Eliminates as-cast dendritic segregation of W and Ta (Group 6 partition)

[3. Structural & Chemical Metrology]
  • Rigaku SmartLab XRD (Cu-K_alpha): Verifies single-phase BCC Bragg peaks (110, 200, 211)
  • JEOL FE-SEM / EDS: Backscattered electron (BSE) mapping confirms chemical homogeneity

[4. Mechanical Hardness & Strength Validation]
  • Vickers Microhardness (Hv, 500 gf load, 15 s dwell, 10x10 indentation grid)
  • Tabor Empirical Verification:  $\sigma_y \approx H_v / 3$ 
  • High-Temperature Uniaxial Compression Testing (up to 1000°C in Ar)
=====
```

5.1 First-Principles DFT Protocol (Tier 3)

To establish quantum ground truth, relaxed SQS supercells will be computed using plane-wave DFT in VASP or Quantum ESPRESSO [79, 80]:

- **PAW Pseudopotentials:** Refractory transition metals are modeled treating semi-core p-states as valence (p-potentials for W, Mo, Ta, Nb, V) [81].
- **Plane-Wave Cutoff:** $E_{\text{cut}} = 520 \text{ eV}$.
- **k-Point Density:** Monkhorst-Pack mesh [82] with a spacing of $\Delta k \leq 0.03 \text{ \AA}^{-1}$.
- **Elastic Stiffness Tensor (C_{ij}):** Small strain tensors ($\pm 1\%$, $\pm 2\%$) applied to the cell yield C_{11} , C_{12} , C_{44} . The Voigt-Reuss-Hill (VRH) shear modulus (G), bulk modulus (B), and Pugh's ductility ratio (B/G) will be computed directly.

5.2 Laboratory Ingot Synthesis (VAR)

Synthesis will be conducted via vacuum non-consumable arc melting on a water-cooled copper hearth:

1. High-purity raw elemental pellets (>99.9%) are weighed according to atomic percentages.
2. The chamber is evacuated to 10^{-4} mbar and backfilled with high-purity Argon.
3. A Titanium getter button is melted first to eliminate trace oxygen and nitrogen.
4. The alloy charge is melted and flipped seven consecutive times to ensure macroscopic chemical homogeneity [41, 42].
5. Ingot buttons undergo homogenization annealing in a high-vacuum tube furnace at 1200°C for 24 hours to eliminate dendritic segregation [43, 44].

5.3 Metrology and Testing

- **Phase Verification:** Rigaku SmartLab X-ray diffraction (Cu-K $_{\alpha}$) will verify the absence of secondary TCP intermetallic reflections [85].
- **Chemical Homogeneity:** Field-emission scanning electron microscopy (FE-SEM) backscattered electron (BSE) imaging and energy-dispersive X-ray spectroscopy (EDS) will quantify elemental partition coefficients.
- **Microhardness:** Vickers microhardness testing (500 gf, 15 s dwell) across a 10×10 matrix will validate yield strength via Tabor's empirical relation: $\sigma_y \approx H_V / 3$ [84].

6. Limitations and Future Work

While MetaForge provides an accelerated computational framework, three physical limitations should be noted:

1. **Pest Oxidation in Refractory Systems:** At intermediate temperatures (600–900°C), unalloyed W and Mo form volatile oxides (MoO $_3$, WO $_3$) [50, 83]. Future iterations will incorporate Aluminum and Chromium additions to promote self-healing $\alpha\text{-Al}_2\text{O}_3$ and Cr $_2\text{O}_3$ passivation scales.
2. **Dynamic Short-Range Order Transitions:** While static SRO is minimized during SQS generation, real alloys exhibit temperature-dependent short-range order that evolves during thermal exposure [32, 33, 37].

3. **CALPHAD Coupling:** Integrating multi-component CALPHAD thermodynamic databases (e.g., TCHEA) will further refine high-temperature solidus and liquidus predictions prior to arc melting [11, 13, 71].
-

7. Conclusion

MetaForge demonstrates an Integrated Computational Materials Engineering framework that bridges high-throughput compositional filtering, physical dislocation mechanics, Monte Carlo SQS annealing, and universal graph neural network potentials. By enforcing the 5-35 at.% Yeh-Cantor boundary and implementing Taylor-Varvenne dislocation physics, the model eliminates unphysical yield strength predictions and avoids artificial structural clustering. The discovered refractory candidate, $\text{W}_{18.5}\text{Mo}_{24.7}\text{Ta}_{26.1}\text{Nb}_{16.8}\text{V}_{13.9}$, exhibits a predicted yield strength of 1.96 GPa at 14.29 g/cm³, supported by thermodynamic entropy stabilization above -63°C. The open-source pipeline provides an accessible foundation for the computational discovery of multi-principal element alloys for extreme engineering applications.

References

1. J. W. Yeh, S. K. Chen, S. J. Lin, J. Y. Gan, T. S. Chin, T. T. Shun, C. H. Tsau, S. Y. Chang, "Nanostructured high-entropy alloys with multiple principal elements: Novel alloy design concepts and outcomes," *Adv. Eng. Mater.* 6 (2004) 299–303.
2. B. Cantor, I. T. H. Chang, P. Knight, A. J. B. Vincent, "Microstructural development in equiatomic multicomponent alloys," *Mater. Sci. Eng. A* 375–377 (2004) 213–218.
3. D. B. Miracle, O. N. Senkov, "A critical review of high entropy alloys and related concepts," *Acta Mater.* 122 (2017) 448–511.
4. E. P. George, D. Raabe, R. O. Ritchie, "High-entropy alloys," *Nat. Rev. Mater.* 4 (2019) 515–534.
5. B. Gludovatz, A. Hohenwarter, D. Catoor, E. H. Chang, E. P. George, R. O. Ritchie, "A fracture-resistant high-entropy alloy for cryogenic applications," *Science* 345 (2014) 1153–1158.
6. F. Otto, A. Dlouhý, C. Somsen, H. Bei, G. Eggeler, E. P. George, "The influences of temperature and microstructure on the tensile properties of a CoCrFeMnNi high-entropy alloy," *Acta Mater.* 61 (2013) 5743–5755.
7. Y. Zhang, T. T. Zuo, Z. Tang, M. C. Gao, K. A. Dahmen, P. K. Liaw, Z. P. Lu, "Microstructures and properties of high-entropy alloys," *Prog. Mater. Sci.* 61 (2014) 1–93.
8. J. W. Yeh, "Recent progress in high-entropy alloys," *Ann. Chim. Sci. Mat.* 31 (2006) 633–648.
9. O. N. Senkov, G. B. Wilks, D. B. Miracle, C. P. Chuang, P. K. Liaw, "Refractory high-entropy alloys," *Intermetallics* 18 (2010) 1758–1765.
10. O. N. Senkov, G. B. Wilks, J. M. Scott, D. B. Miracle, "Mechanical properties of Nb₂₅Mo₂₅Ta₂₅W₂₅ and V₂₀Nb₂₀Mo₂₀Ta₂₀W₂₀ refractory high entropy alloys," *Intermetallics* 19 (2011) 698–706.
11. A. Takeuchi, A. Inoue, "Classification of bulk metallic glasses by atomic size difference, heat of mixing and period of constituents," *Mater. Trans.* 46 (2005) 2817–2829.

12. A. R. Miedema, F. R. de Boer, R. Boom, "Model predictions for the enthalpy of formation of transition metal alloys," **CALPHAD** 1 (1977) 341–359.
13. X. Yang, Y. Zhang, "Prediction of high-entropy alloys properties: First-principles calculation and thermodynamic prediction," **Mater. Chem. Phys.** 132 (2012) 233–238.
14. S. Guo, C. Ng, J. Lu, C. T. Liu, "Effect of valence electron concentration on stability of fcc or bcc phase in high entropy alloys," **J. Appl. Phys.** 109 (2011) 103505.
15. Y. Zhang, Y. J. Zhou, J. P. Lin, G. L. Chen, P. K. Liaw, "Solid-solution phase formation rules for multi-component alloys," **Adv. Eng. Mater.** 10 (2008) 534–538.
16. M. C. Tropicovsky, J. R. Morris, P. R. C. Kent, A. R. Lupini, G. M. Stocks, "Criteria for predicting the formation of single-phase high-entropy alloys," **Phys. Rev. X** 5 (2015) 011041.
17. M. G. Poletti, L. Battezzati, "Electronic and thermodynamic criteria for the occurrence of high entropy alloys in binary systems," **Acta Mater.** 75 (2014) 297–306.
18. D. J. M. King, S. C. Middleburgh, A. G. McGregor, M. B. Cortie, "Predicting the formation and stability of single phase high-entropy alloys," **Acta Mater.** 113 (2016) 230–245.
19. O. N. Senkov, J. D. Miller, D. B. Miracle, C. Woodward, "Accelerated exploration of multi-principal element alloys with solid solution phases," **Nat. Commun.** 6 (2015) 6529.
20. L. J. Santodonato, Y. Zhang, M. Feygenson, C. M. Parish, M. C. Gao, R. J. K. Weber, J. C. Neugefeind, Z. Tang, P. K. Liaw, "Deviation from high-entropy configurations in the atomic distributions of a model complex-concentrated alloy," **Nat. Commun.** 6 (2015) 5964.
21. C. Varvenne, A. Luque, W. A. Curtin, "Theory of strengthening in dilute and concentrated solid-solution alloys," **Acta Mater.** 118 (2016) 164–176.
22. F. Maresca, W. A. Curtin, "Mechanistic origin of high strength in refractory BCC high entropy alloys up to 1900K," **Acta Mater.** 182 (2020) 144–154.
23. F. Maresca, W. A. Curtin, "Theory of screw dislocation strengthening in random BCC alloys from dilute to 'High-Entropy' alloys," **Acta Mater.** 182 (2020) 235–249.
24. W. A. Curtin, D. L. Olmsted, L. G. Hector, "A predictive mechanism for dynamic strain aging in aluminum-magnesium alloys," **Nat. Mater.** 5 (2006) 875–880.
25. R. L. Fleischer, "Solution hardening by lattice distortions: Large size effects," **Acta Metall.** 9 (1961) 996–1000.
26. R. Labusch, "A statistical theory of solid solution hardening," **Phys. Status Solidi** 41 (1970) 659–669.
27. G. I. Taylor, "Plastic strain in metals," **J. Inst. Met.** 62 (1938) 307–324.
28. I. Toda-Caraballo, P. E. J. Rivera-Díaz-del-Castillo, "Modelling solid solution hardening in high entropy alloys," **Acta Mater.** 85 (2015) 14–23.
29. B. Gwalani, S. Soni, D. Choudhuri, M. Lee, J. Y. Hwang, S. J. Ryu, R. Banerjee, "Stability of ordered L12 and B2 precipitates in a CoCrFeNiAl medium-entropy alloy," **Acta Mater.** 196 (2020) 221–230.
30. G. Laplanche, A. Kostka, O. M. Horst, G. Eggeler, E. P. George, "Microstructure evolution and critical stress for twin nucleation in the CrMnFeCoNi high-entropy alloy," **Acta Mater.** 118 (2016) 152–163.

31. A. Zunger, S. H. Wei, L. G. Ferreira, J. E. Bernard, "Special quasirandom structures," **Phys. Rev. Lett.** 65 (1990) 353–356.
32. Q. Ding, Y. Zhang, X. Chen, X. Fu, D. Chen, S. Chen, L. Gu, F. Wei, H. Bei, R. Gao, M. Chen, J. Li, Z. Zhang, T. Zhu, R. O. Ritchie, Q. Yu, "Tuning element distribution, chemical short-range order and properties in multi-principal element alloys," **Nature** 574 (2019) 223–227.
33. R. Zhang, S. Zhao, J. Ding, Y. Chong, T. Jia, C. Ophus, M. Asta, R. O. Ritchie, A. M. Minor, "Short-range order and its impact on the CrCoNi medium-entropy alloy," **Nature** 581 (2020) 283–287.
34. J. M. Cowley, "An approximate theory of order in alloys," **Phys. Rev.** 77 (1950) 669–670.
35. B. E. Warren, **X-Ray Diffraction**, Addison-Wesley, Reading, MA, 1969.
36. D. Chen, S. Zhao, R. Zhang, Z. Mao, M. Asta, R. O. Ritchie, "Direct observation of chemical short-range order in a medium-entropy alloy," **Proc. Natl. Acad. Sci. USA** 118 (2021) e2020540118.
37. N. V. Kostichenko, F. Körmann, J. Neugebauer, A. Shapeev, "Impact of chemical short-range order on the mechanical properties of MoNbTaW," **Phys. Rev. Mater.** 3 (2019) 024408.
38. Q. J. Li, H. Sheng, E. Ma, "Strengthening in multi-principal element alloys with local chemical order," **Nat. Mater.** 18 (2019) 700–705.
39. M. Widom, W. P. Huhn, S. Maiti, W. Steurer, "Hybrid Monte Carlo/molecular dynamics simulation of high-entropy alloys," **Phys. Rev. B** 89 (2014) 174202.
40. A. Tamm, A. Aabloo, M. Klintonberg, M. Stocks, A. Caro, "Atomic-scale properties of Ni-based high-entropy alloys," **Phys. Rev. Lett.** 114 (2015) 165502.
41. O. N. Senkov, S. V. Senkova, C. Woodward, "Effect of Al on the microstructure and properties of two refractory high-entropy alloys," **Acta Mater.** 68 (2014) 214–228.
42. D. B. Miracle, O. N. Senkov, J. P. Couzinié, "Refractory high-entropy alloys: A review," **J. Mater. Res.** 35 (2020) 843–860.
43. J. K. Jensen, B. A. Welk, R. E. A. Williams, J. M. Jensen, G. B. Viswanathan, M. A. Gibson, H. L. Fraser, "Characterization of the microstructure and mechanical properties of a MoNbTaW refractory high entropy alloy," **Scr. Mater.** 121 (2016) 1–5.
44. J. P. Couzinié, O. N. Senkov, D. B. Miracle, G. Dirras, "Comprehensive data on the microstructure and mechanical properties of refractory high entropy alloys," **Mater. Lett.** 213 (2018) 245–248.
45. L. Liliensten, J. P. Couzinié, L. Perrière, A. Hocini, M. Veron, G. Dirras, "Study of a TiNbZrMo high-entropy alloy for biomedical applications," **Acta Mater.** 132 (2017) 69–78.
46. S. Sheikh, S. Shafeie, Q. Hu, J. Ahlström, C. Persson, J. Vestman, S. Guo, "Alloy design for intrinsically ductile refractory high-entropy alloys," **Intermetallics** 77 (2016) 63–74.
47. Z. D. Han, N. Chen, S. F. Zhao, L. W. Fan, G. N. Yang, Y. Shao, K. F. Yao, "Effect of Ti addition on mechanical properties of NbMoTaW refractory high entropy alloy," **Intermetallics** 93 (2018) 136–142.
48. B. Gorr, F. Müller, M. Azim, H. J. Christ, T. Müller, H. Chen, A. Kauffmann, M. Heilmaier, "High-temperature oxidation behavior of refractory high-entropy alloys," **J. Alloys Compd.** 688 (2016) 468–477.

49. B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, G. Ceder, "CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling," **Nat. Mach. Intell.** 5 (2023) 1031–1041.
50. C. Chen, S. P. Ong, "A universal graph deep learning interatomic potential for the periodic table," **Nat. Comput. Sci.** 2 (2022) 718–728.
51. I. Batatia, D. P. Kovacs, G. N. Simm, C. Ortner, G. Csányi, "MACE: Higher order equivariant and message passing interatomic potentials," **Adv. Neural Inf. Process. Syst.** 35 (2022) 11423–11436.
52. T. Xie, J. C. Grossman, "Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties," **Phys. Rev. Lett.** 120 (2018) 145301.
53. A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, W. G. Cheon, E. D. Cubuk, "Scaling deep learning for materials discovery," **Nature** 624 (2023) 80–85.
54. J. Behler, M. Parrinello, "Generalized neural-network representation of high-dimensional potential-energy surfaces," **Phys. Rev. Lett.** 98 (2007) 146401.
55. A. P. Bartók, M. C. Payne, R. Kondor, G. Csányi, "Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons," **Phys. Rev. Lett.** 104 (2010) 136403.
56. A. P. Thompson, L. P. Swiler, C. R. Trott, S. M. Foiles, G. J. Tucker, "Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials," **J. Comput. Phys.** 285 (2015) 316–330.
57. S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, B. Kozinsky, "E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials," **Nat. Commun.** 13 (2022) 2453.
58. K. T. Schütt, H. E. Sauceda, P. J. Kindermans, A. Tkatchenko, K. R. Müller, "SchNet – A deep learning architecture for molecules and materials," **J. Chem. Phys.** 148 (2018) 241722.
59. L. Ward, A. Agrawal, A. Choudhary, C. Wolverton, "A general-purpose machine learning framework for predicting properties of inorganic materials," **npj Comput. Mater.** 2 (2016) 16028.
60. L. Ward, A. Dunn, A. Faghaninia, N. E. R. Zimmermann, S. Bajaj, Q. Wang, J. Montoya, J. Zheng, N. Voorhis, K. A. Persson, "Matminer: An open-source toolkit for materials data mining," **Comput. Mater. Sci.** 152 (2018) 60–69.
61. A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, K. A. Persson, "Commentary: The Materials Project: A materials genome approach to accelerating materials innovation," **APL Mater.** 1 (2013) 011002.
62. S. Curtarolo, W. Setyawan, G. L. Hart, M. Jahnatek, R. V. Chepulskii, R. H. Taylor, S. Wang, J. Xue, K. Yang, O. Levy, "AFLOW: An automatic framework for high-throughput materials discovery," **Comput. Mater. Sci.** 58 (2012) 218–226.
63. J. E. Saal, S. Kirklin, M. Aykol, B. Meredig, C. Wolverton, "The Open Quantum Materials Database (OQMD): Assessing the accuracy of DFT formation energies," **JOM** 65 (2013) 1501–1509.
64. O. Isayev, C. Oses, C. Toher, E. Gossett, S. Curtarolo, A. Tropsha, "Universal fragment descriptors for predicting properties of inorganic crystals," **Nat. Commun.** 8 (2017) 15679.
65. S. P. Ong, W. D. Richards, A. Jain, G. Hautier, M. Kocher, S. Cholia, D. Gunter, V. L. Chevrier, K. A. Persson, G. Ceder, "Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis," **Comput.*

- Mater. Sci.* 68 (2013) 314–319.
66. R. Ramprasad, R. Batra, G. Pilania, A. Mannodi-Kanakkithodi, C. Kim, "Machine learning in materials informatics: Recent applications and prospects," **npj Comput. Mater.** 3 (2017) 54.
 67. K. T. Butler, D. W. Davies, H. Cartwright, O. Isayev, A. Walsh, "Machine learning for molecular and materials science," **Nature** 559 (2018) 547–555.
 68. J. Schmidt, M. R. G. Marques, S. Botti, M. A. L. Marques, "Recent advances and applications of machine learning in solid-state materials science," **npj Comput. Mater.** 5 (2019) 83.
 69. Z. Rao, P. Y. Tung, R. Lu, Y. Wei, A. Ferrari, A. Xie, E. J. George, D. Raabe, Z. Li, "Machine learning-guided discovery of high-entropy alloys with superior properties," **Science** 378 (2022) 78–85.
 70. T. Lookman, P. V. Balachandran, D. Xue, R. Yuan, "Active learning in materials science with emphasis on adaptive design," **npj Comput. Mater.** 5 (2019) 21.
 71. J. M. Rickman, T. Lookman, S. V. Kalinin, "Materials informatics: An emerging paradigm for materials design," **Nat. Commun.** 10 (2019) 2618.
 72. K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," **IEEE Trans. Evol. Comput.** 6 (2002) 182–197.
 73. R. LeSar, **Introduction to Computational Materials Science: Fundamentals to Applications**, Cambridge University Press, Cambridge, 2014.
 74. D. Xue, P. V. Balachandran, J. Hogden, J. Theiler, D. Q. Xue, T. Lookman, "Accelerated search for materials with targeted properties by adaptive design," **Nat. Commun.** 7 (2016) 11241.
 75. C. Wen, Y. Zhang, C. Wang, D. Xue, T. Lookman, Y. Su, "Machine learning assisted design of high entropy alloys with desired property," **Acta Mater.** 170 (2019) 109–117.
 76. B. L. DeCost, M. D. Callahan, M. A. Tschopp, "Exploring the composition-structure-property landscape of high entropy alloys with active learning," **Acta Mater.** 165 (2019) 595–605.
 77. W. Kohn, L. J. Sham, "Self-consistent equations including exchange and correlation effects," **Phys. Rev.** 140 (1965) A1133–A1138.
 78. J. P. Perdew, K. Burke, M. Ernzerhof, "Generalized gradient approximation made simple," **Phys. Rev. Lett.** 77 (1996) 3865–3868.
 79. P. E. Blöchl, "Projector augmented-wave method," **Phys. Rev. B** 50 (1994) 17953–17979.
 80. H. J. Monkhorst, J. D. Pack, "Special points for Brillouin-zone integrations," **Phys. Rev. B** 13 (1976) 5188–5192.
 81. O. N. Senkov, S. V. Senkova, D. M. Dimiduk, C. Woodward, D. B. Miracle, "Oxidation behavior of a refractory high entropy alloy Al_{0.4}Hf_{0.6}NbTaTiZr," **NPJ Mater. Degrad.** 2 (2018) 43.
 82. D. Tabor, **The Hardness of Metals**, Oxford University Press, Oxford, 1951.
 83. H. M. Rietveld, "A profile refinement method for nuclear and magnetic structures," **J. Appl. Crystallogr.** 2 (1969) 65–71.

84. M. J. S. Spencer, T. Lookman, *Materials Informatics: Methods, Tools and Applications*, Wiley-VCH, Weinheim, 2020.
85. E. P. George, W. A. Curtin, C. C. Tasan, "High entropy alloys: A focused review of mechanical properties and deformation mechanisms," *Acta Mater.* 188 (2020) 435–474.